MILITARY SPECIFICATION

RADOME CW-364/DPN-24

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be a radome for the protection of the Antenna of the Target Seeker Group OA-831/DPN-24, as an integral section of an AAM-N-6 air-launched missile.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Unit	Type Designation	Max. Allow. Size (Inches)	Max. Allow. Wt. Lbs.	Applicable Paragraph
Radome	CW-364/DPN-24	8.01 diameter (base) 19.75 length	6.0	3.3

1.3 Accessories - The equipment covered by this specification shall operate with the following associated equipment; however, this associated equipment is not to be supplied as a part of this equipment.

Target Seeker Group OA-831/DPN-24

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on date of invitation for bids shall form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-7705 Radome; General Specification

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MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment, Assemblies and Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection; Format for

Bureau of Aeronautics

SD-3006-1-1	Detail Specification for Model AAM-N-6 Guided Missile

2.2 Availability of Documents -

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attn: Code ODPT.

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attn: Electronics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Electronics Division, Bureau of Aeronautics.

(3) Referenced Specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 General - The radomes shall be fabricated in accordance with the materials and workmanship requirements of Specification MIL-R-7705.

3.1.1 Materials and Fabrication Techniques - In the selection of materials and fabrication techniques, fulfillment of major design objectives and meeting of environmental requirements shall be of prime consideration. Features of materials and fabrication techniques contained in the referenced government specifications shall not prescribe, nor limit, the consideration and use of materials and fabrication techniques which may be demonstrably superior for the intended purpose.

3.2 Workmanship - Workmanship shall be in accordance with the highest grade commercial practice.

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3.2.1 Surfaces - The surface shall be smooth and uniform in finish and appearance, to the maximum degree commensurate with electrical requirements.

3.2.2 Surface Fillers - Resin or putty shall not be used as a surface filler. This requirements shall not be construed to prohibit the use of resins in the major processed structure of the radomes.

3.3 Performance - All radomes shall meet the following performance requirement

3.3.1 Electrical - In lieu of the electrical requirements of MIL-R-7705 throughout the operating frequency band for a guided missile of the AAM-N-6 type:

3.3.1.1 Attenuation - The radome shall not attenuate the received radar signal by more than 30% when the Target Seeker Antenna is being gimbaled over its range (± 46° in pitch and in yaw.

3.3.1.2 Reflection - Power reflection requirements for the radome shall be commensurate with the other electrical requirements specified herein.

3.3.1.3 Antenna Pattern Distortion -

3.3.1.3.1 Side-Lobes - Side-lobes of the Target Seeker Antenna pattern shall not be increased more than 3 db by the presence of the radome.

3.3.1.3.2 Ambiguity - The ambiguity (information sense reversal) of the Target Seeker shall not be moved by more than 1° toward the boresight axis by the presence of the radome.

3.3.1.4 Antenna Beam Deflection - For the purposes of this specification, the following definitions shall apply to the requirements for Antenna Beam Deflection:

(1) Boresight Error - With the Target Seeker Antenna on boresight, the apparent received radar error signal (indicating target displacement), produced by the presence of the radome is defined as the "boresight error". The error introduced by the radome is taken as positive when it indicates a target displacement in the opposite direction to the gimbaled motion of the radome about the Target Seeker Antenna.

(2) Boresight Error Slope - The rate of change of the boresight error in yaw with respect to the gimbal angular displacement of the radome in yaw is defined as the "boresight error slope". This "error slope" is taken as positive when the rate of change is in the opposite direction to the motion of the radome about the Target Seeker Antenna.

(3) Crosstalk Error Slope - The rate of change of the boresight error in pitch with respect to the gimbal angular displacement of the radome in yaw is defined as the "crosstalk error slope".

3.3.1.4.1 Boresight Error - Over the gimbal range (± 46° in pitch and yaw), the radome shall not introduce a boresight error of more than 1°.

3.3.1.4.2 Boresight Error Slope - Over the gimbal range, the radome shall not introduce a negative boresight error slope of more than 0.07° error per degree gimbal; not positive boresight error slope of more than 0.08° error per degree gimbal. Only boresight error slopes (either positive or negative) which persist for more than 3° of radome gimbal shall be considered applicable to this requirement.

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3.3.1.4.3 Crosstalk Error Slope - Over the gimbal range, the radome shall not introduce a crosstalk error slope of more than 0.05° per degree gimbal. Only crosstalk error slopes which persist for more than 3° of radome gimbal shall be considered applicable to this requirement.

3.3.1.5 Dielectric Constant - The dielectric constant for the final processed radome material shall be 5.3 to 6.3.

3.3.1.6 Interchangeability - Radomes produced under this specification shall be interchangeable with one another on the latest produced models of the Target Seeker Group OA-831/DPN-24.

3.3.2 Environmental (Service Conditions) - The equipment as a part of a missile of the AAM-N-6 type shall operate satisfactorily under any of the service conditions, or reasonable combinations of service conditions as specified in SD-3006-1-1, subject to the following provision: during storage, standby, or prelaunch flight, the radome shall be protected against surface moisture by a close-fitting plastic sleeve which shall be removed automatically when the AAM-N-6 missile is launched.

3.3.3 Structural

3.3.3.1 Strength and Fatigue - The radome airframe joint shall withstand a shear load of 1,150 pounds and a corresponding bending moment of 11,025 inch-pounds without failure at an outside surface temperature of 430° F.

3.3.3.2 Gun Blast - There are no requirements for resistance to gun blast for this radome.

3.4 Construction -

3.4.1 Dimensions - The dimensions, shape, and other details of construction shall be as shown in Figure 1 of this specification.

3.4.2 Standard Parts - There shall be no requirement for JAN or AN standard parts for this radome.

3.4.3 Paint - There shall be no requirement for paint for this radome. Paint used for marking the radome shall not impair the performance characteristics as outlined in this specification.

3.4.4 Inserts - There shall be no requirement for inserts for this radome other than the plastic thread ring at the base.

3.4.5 Metallic Materials - Metallic powders may be used in the processing of the radome material; however, neither materials containing metallic impurities nor processes permitting the introduction of metallic impurities, shall be employed in the construction of the radome. A metal tip of stainless steel shall be used to provide protection against rain erosion.

3.4.6 Installation - The radome shall be constructed to enable installation, or removal, from the missile in approximately two minutes. Suitable indexing shall be provided to ensure the correct polarization mounting for the radome.

3.4.7 Serviceability - The radome shall be constructed to provide easy access for service and repair of that section of the Target Seeker Group which it encloses.

3.4.8 Life - The radome shall be constructed to provide long life under any reasonable combination of severe conditions of service use.

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3.4.9 Sealing Provisions - An "O ring" type seal, or other suitable mounting arrangement shall be provided to protect that section of the Target Seeker Group which the radome encloses against salt spray, sand and dust, etc.

3.5 Nomenclature and Nameplates - Nomenclature assignment shall be in accordance with the applicable requirements of Specification MIL-N-18307. Identification markings shall be in accordance with paragraph 3.6 herein.

3.6 Marking - The following information shall be marked on all radomes after the final finishing process:

3.6.1 Outside - In contrasting colors after final finish is applied as close as possible to mounting end:

(1) "DO NOT PAINT"

(2) "RADOME OFF"

(3) Radome Serial Number

(4) An arrow to indicate "up air launch"

3.6.2 Inside - In contrasting color circumferentially around the radome at the base of its major thickness as close as possible to thread ring.

(1) Nomenclature

(2) Drawing Number

(3) Detail Specification Number

(4) Contract Number

(5) Manufacturer's Serial Number

(6) Mold Number (If used)

(7) Manufacturer's Name

(8) Index Mark (plane of polarization)

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4. QUALITY ASSURANCE PROVISIONS

4.1 General Procedures and Methods - The procedures and methods for conducting tests shall be as specified in section 4 of the Detail Specification SD-3006-1-1, for the AAM-N-6 Guided Missile.

5. PREPARATION FOR DELIVERY

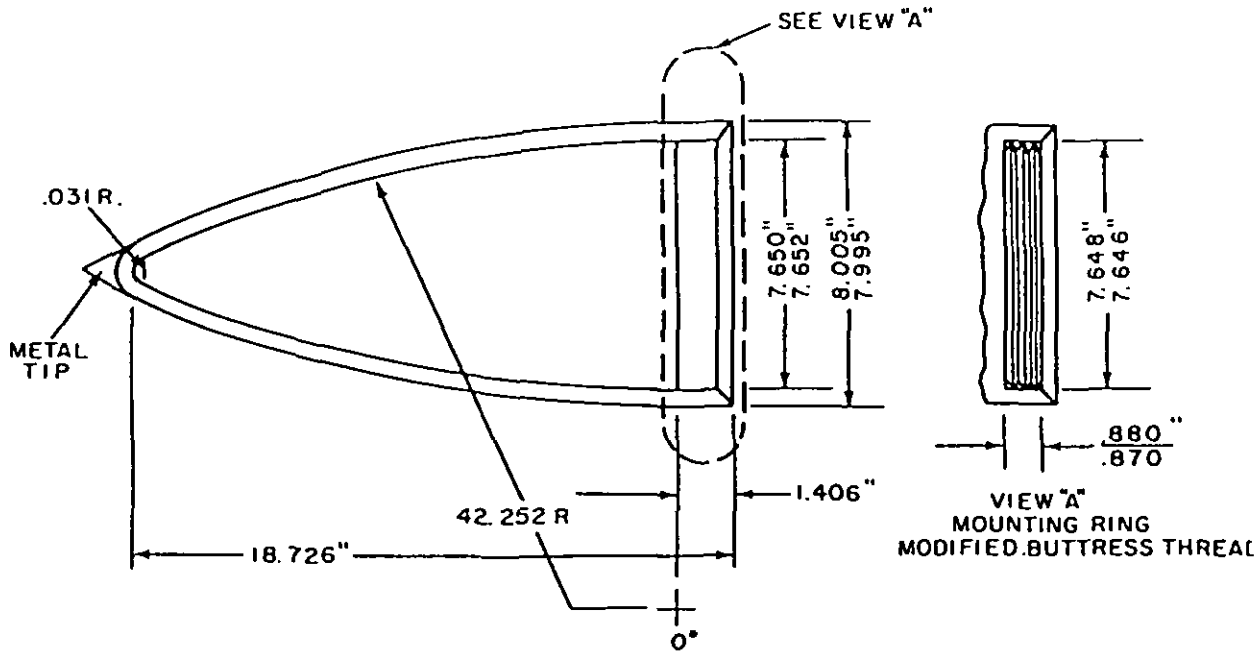
5.1 General - Radomes and spare radomes shall be packed as specified in the contract.

6. NOTES

(Not applicable)

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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NOTES:

1. Markings shall be in accordance with section 3.6 of this specification.
2. Materials:
   a. The solid wall radome shall consist of glass cloth impregnated with the following - Stypol 16B, Titanium Dioxide, Tertiary Butyl Hydroperoxide, Luperco, ATC paste.
   b. The mounting ring shall consist of glass cloth, wrappings of glass cord, and epoxy resin.
   c. The nose cap (metal tip) shall consist of stainless steel.
3. The mounting ring and solid wall radome shall be bonded together, and the metal tip shall be held in place by a suitable retaining pin.
4. Radome wall thickness shall vary in accordance with the measured dielectric constant.

Figure 1 - Radome - CW-364/DPN-24

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